Curriculum Vitae

Gustavo Marques de Oliveira Martins

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Personal Information

Name: Gustavo Marques de Oliveira Martins
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Address

IMPA - Instituto Nacional de Matemática Pura e Aplicada
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Education

2023-2027 (expected) **Mathematics Ph.D.**
IMPA - Instituto de Matemática Pura e Aplicada, Rio de Janeiro/RJ Brazil
Advisor: Lucas Ambrozio
Visiting Student at University of Chicago (2025-2026)
Host: André Neves
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)/ CAPES Programa Doutorado-sanduíche no Exterior (PDSE)

2021-2023 **Mathematics M.S.**
IMPA - Instituto de Matemática Pura e Aplicada, Rio de Janeiro/RJ Brazil
Advisor: Lucas Ambrozio
Thesis: A brief tour through the theory of Zoll surfaces
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)/ Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro (FAPERJ)

2017-2022 **Applied Mathematics B.S. - Emphasis in Business Mathematics**
UFRJ - Universidade Federal do Rio de Janeiro/RJ Brazil

Research interests

Mathematics
Differential Geometry
Geometric Analysis

Publications and Preprints

Gustavo Martins, *Topological and spectral rigidity of hypersurface Zoll manifolds*, preprint, arXiv:2609.20689 [math.DG].

Invited talks

10/2026 Differential Geometry Seminar – IMPA
Title: Topological and spectral rigidity of hypersurface Zoll manifolds

General talks

2021	XLII Jornada Giulio Massarani de Iniciação Científica, Tecnológica, Artística e Cultural (JICTAC 2020): "O Teorema de Mergulho de Kodaira", UFRJ, Brazil.
2021	XLII Jornada Giulio Massarani de Iniciação Científica, Tecnológica, Artística e Cultural (JICTAC 2020):"Suavização de tábuas de mortalidade via modelos dinâmicos lineares", UFRJ, Brazil.
2019	Encontros Matemáticos: "Geometria e Economia" – UFRJ, Brazil.

Events and Conferences

2026	XXII Escola Brasileira de Geometria Diferencial (participant).
2026	2026 Lake Michigan Workshop on Geometric Analysis Seminar (participant)
2025	35º Colóquio Brasileiro de Matemática (participant).
2025	Problem-solving Workshop in Computational Geometric Analysis (participant)
2025	Systolic and Diastolic Geometry (participant).
2024	Special Holonomy and Geometric Structures on Complex Manifolds (participant)
2023	34º Colóquio Brasileiro de Matemática (participant).
2022	Conference IMPA 70 years (participant).
2021	33º Colóquio Brasileiro de Matemática (participant).
2020	3ª Semana da Engenharia Matemática e Matemática Aplicada da UFRJ (participant).
2019	32º Colóquio Brasileiro de Matemática (participant).

Teaching Experience

IMPA	Teaching Assistant Partial Differential Equations	2023
UFRJ	Teaching Assistant (ran problem sessions), MAE240, Real Analysis.	2019.2
Pontifícia Universidade Católica (PUC-Rio)	Teaching Assistant (ran problem sessions), ECO1721, Introduction to Economic Statistics.	2018.2
	Teaching Assistant (ran problem sessions), ECO1109, Introduction to Economics I.	2018.1

Students and mentoring

University of Chicago	Directed Reading Program (DRP) mentor Student mentored: Josh Johnson Topic: Minimal surfaces and the Plateau Problem Description: The goal was to study the solution to the Plateau Problem by Douglas and Rado as explained in Chapter 4 of T. H. Colding and W. P. Minicozzi II's book *A Course in Minimal Surfaces*.	2026

Honours and Awards

2025	Bolsa CAPES do Programa Doutorado-sanduíche no Exterior (PDSE).
2022	Bolsa Nota 10, FAPERJ.
2021	Best project presented in the Mathematics field at XLII Jornada Giulio Massarani de Iniciação Científica, Tecnológica, e Cultural (JICTAC 2020 - Edição Especial).

Other Activities

2020-2021	Iniciação Científica, UFRJ Advisors: Mariane Branco Alves, Thais Cristina Oliveira da Fonseca, Viviana das Graças Ribeiro Lobo Description: the project was done at the Laboratório de Matemática Aplicada (LabMA) with funding from Mongeral Aegon. The goal was to apply Dynamical Linear Models in the study and smoothing of actuarial life tables.
2018-2021	Iniciação Científica, UFRJ Advisor: Andrew Clarke Description: the project was intended as an introduction to differential and complex geometry, with an emphasis in understanding the Kodaira Embedding Theorem.
2017-2018	Attended classes in the Economics Department at PUC-Rio. This includes courses in Micro and Macroeconomics, Public Sector Economics, Monetary Economics and Econometrics.